When to Use

To compare the means of two quantitative variables obtained from dependent samples (repeated measures or matched groups). The two scores might be the same variable measured at two different times or under two different conditions, two comparable variables measured at the same time, or some combination.

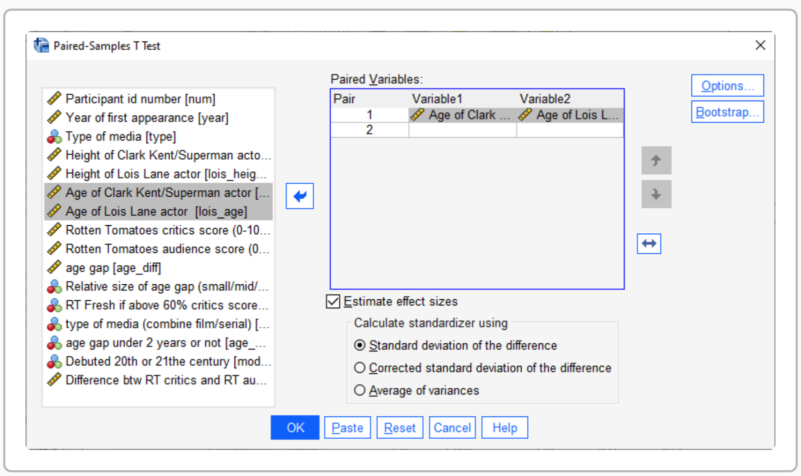
Research Hypothesis: Actors playing Clark Kent are older on average than actors playing Lois Lane, reflecting traditional casting preferences for male-female romantic pairings.

Null Hypothesis: The mean ages of actors playing Clark Kent and Lois Lane in Superman portrayals are equal.

Running a Paired-Samples T-Test

Analyze → Compare Means → Paired-Samples T-Test

- Highlight the **two** variables that are the DV for the two IV conditions and click the arrow



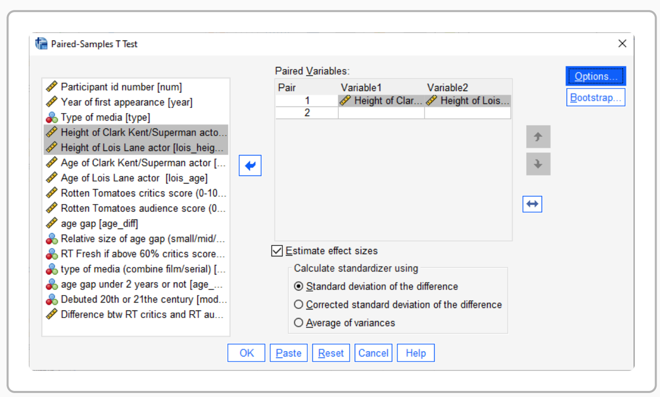
SPSS Syntax

```
T-TEST
PAIRS=clark_age WITH
lois_age (PAIRED). 1
```

1 List variable pairs: first variable WITH second variable, wrapped in (PAIRED) to indicate a repeated-measures design

You can enter multiple pairs in one command. For example, to also test height:

```
T-TEST PAIRS=clark_age clark_height_in WITH lois_age
lois_height_in (PAIRED).
```



SPSS Output

Descriptive Statistics

	N	Mean	Std. Deviation	Std. Error	Minimum	Maximum
Actor age: Clark Kent	9	30.58	4.88	1.63	24.47	37.84
Actor age: Lois Lane	9	30.17	6.41	2.14	23.32	40.10
Valid N (listwise)	9					

The p-value of 0.873 means that there is about a 87.3% chance that this result is a Type I error.

Remember, even if the printout shows p = .000, never report p = .000, because that would sug-

Paired Samples Test

	Mean	Std. Deviation	Std. Error Mean	t	df	Sig. (2-tailed)
clark_age - lois_age	0.402	7.331	2.444	0.165	8	.873

gest there is no possibility of a Type 1 error. Instead, report “p < .001”.

Why a correlation for a t-test?

Data from a repeated measures design can be used to ask two different types of research questions. The paired (repeated measures, within-group, within-subject) t-test is used to ask if there is a significant difference between the means of the repeated measures. Correlation can be used to ask if there is a linear relationship between the repeated measures.

Be sure you are looking at the right statistic for the specific research question or hypothesis you have!!!

Reporting the Results

Sample APA Write-Up

The mean ages of actors playing Clark Kent and Lois Lane across 9 Superman portrayals are summarized in the table above. Actors playing Clark Kent Clark Kent (M = 30.58, SD = 4.88) were compared with actors playing Lois Lane Lois Lane (M = 30.17, SD = 6.41) using a paired-samples t-test. There was no significant difference between the two conditions, $t(8) = 0.16$, $p = 0.873$, with a mean difference of 0.40 years. Contrary to the research hypothesis, there was no significant difference in the mean ages of actors playing Clark Kent and Lois Lane.

It is important to report the univariate statistics for the dependent variable for both groups before presenting the ANOVA results. Often these are presented in a table or a figure.

As in the example, be sure to communicate:

- The research hypothesis (if there is one)
- The statistical results
- Whether or not those results support the research hypothesis

Sometimes, the univariate statistics are presented in text, along with the correlation results.